EXB-08 · EXAM BLUEPRINT

Sample items with rationale

Sixteen original study items across all thirteen domains, with every option explained and every calculation shown

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

Summary

Sixteen original sample items covering all thirteen PCL-AI domains, written to the published blueprint in the examination's format and cognitive levels. Every item carries a full rationale that explains why the key is the key and what specific practitioner error each distractor represents, and every calculation shows its substitution. A closing section reads several of the items as an item writer would, so that candidates can see what a well-constructed item is doing to them. These are study items, written for publication and held separately from the live examination bank; no live examination content is ever published. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

About this document

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Contents

Sample items with rationale

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Who this is for. Candidates preparing for the PCL-AI; training providers building practice material; and anyone who wants to see what the Institute means when it says its items assess judgement rather than recall.

1. What these items are, and what they are not

They are study material. Every item below was written for publication. They are authored to the same blueprint, the same four-option single-best-answer format and the same cognitive levels as the examination, which is what makes them useful preparation.

They are held apart from the live bank, permanently. No item published here is or will become a scored item, and no scored item is or will be published here. That separation is absolute and it runs in both directions — see `EXB-07 — Examination security and integrity` §2.1. A certification body that publishes its live items has converted its examination into a memory test, and one that promotes its published items into the bank has published its own answer key.

No live examination content is ever published, in a sample paper, a course, a marketing document or on request.

The number of items here is not an item count. Sixteen is how many fit usefully in a published document. **The PCL-AI has no published item count**; item counts will be confirmed by a formal job-task analysis — see `EXB-06 — Job-task analysis: where the blueprint gets its authority`.

The mix here is not the form's specification either. These sixteen are distributed for domain coverage — one item in each of the twelve domains outside the AI group, and four in Domain 13 — so that a candidate can see the whole frame. A live form is assembled to the published 40/40/20 group weighting described in `EXB-02 — Domain weightings and the 40/40/20 rule`, which is a different distribution.

2. How to use them

Work each item completely before reading its rationale. Write down your answer and, more importantly, write down the intermediate figure or the position you established on the way to it. Then read the rationale in full, including the options you did not choose.

The rationale is the part that teaches. Getting an item right for the wrong reason is worth almost nothing in preparation, and the only way to detect it is to compare your reasoning against the explanation rather than your letter against the key.

When you get one wrong, identify **which** distractor you chose and **which named error** it corresponds to. Every distractor below was built from a specific mistake practitioners actually make. Knowing which one you made is the diagnostic; knowing that you scored twelve out of sixteen is not.

3. Reading the tags

Each item is tagged `[Domain · Knowledge area · Cognitive level]` — for example `[D6 · KA 6.3 · Analysis]`. The cognitive levels are defined in `EXB-03 — Cognitive levels and item types` §2.

Across these sixteen items: two are Recall, five are Application and nine are Analysis. That is broadly the shape the examination intends — application-weighted, with Recall a genuine minority — though the exact proportions on a live form form part of the examination specification and are not published as a fixed mix.

All monetary figures are illustrative and expressed in USD. All periods, rates and quantities are illustrative. No item below describes a real project, organisation or person.

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4. Group 1 — Project accounting & finance (Domains 1–4)

Item 1 `[D1 · KA 1.3 · Application]`

Illustrative figures.

At the 31 March cut-off, a subcontractor has completed work on site independently valued at USD 180,000. The subcontractor has submitted invoices totalling USD 120,000. The cost ledger records invoiced amounts only. What treatment does the March cost position require?

- **A.** Accrue USD 60,000 as cost incurred and not yet invoiced.
- **B.** Make no entry: the cost is recognised when the subcontractor's invoice is received.
- **C.** Accrue USD 180,000 in addition to the USD 120,000 already recorded.
- **D.** Recognise a provision of USD 60,000 for a liability of uncertain amount.

Key: A.

Work performed in a period is a cost of that period, whatever the invoicing has done. The accrual is the gap between value of work executed and value invoiced:

```
180,000 − 120,000 = 60,000
```

B is the cash-basis instinct, and it is the single most common cause of a project that appears to be under-running at every cut-off and over-running at the final account. It defers cost into the wrong period and flatters the current one.

C double-counts. Accruing the full 180,000 on top of the 120,000 already in the ledger records 300,000 of cost for 180,000 of work.

D confuses two adjacent concepts. A provision is recognised for a liability of uncertain timing or amount. Here the work is done, the value is measured and the obligation is certain — that is an accrual. Getting this boundary wrong is not merely presentational: provisions attract different disclosure and different scrutiny.

Item 2 `[D2 · KA 2.2 · Application]`

Illustrative figures.

A contractor recognises revenue over time on a construction contract using a cost-to-cost input method. The transaction price is USD 10,000,000 and the estimated total cost to complete is USD 8,000,000. Costs incurred to date are USD 3,000,000. What revenue should be recognised to date?

- **A.** USD 5,000,000
- **B.** USD 3,750,000
- **C.** USD 3,000,000
- **D.** USD 2,400,000

Key: B.

Progress is measured as costs incurred against estimated total costs, and that proportion is applied to the transaction price:

`Progress = 3,000,000 ÷ 8,000,000 = 0.375 = 37.5 % Revenue to date = 0.375 × 10,000,000 = 3,750,000`

A assumes the contract is half complete because it feels half complete, ignoring the measured progress figure the stem supplies. The method is called an input method precisely because progress is measured, not estimated by impression.

C recognises revenue equal to cost incurred — a zero-margin position. That treatment exists in specific circumstances, but it is not the general cost-to-cost result, and applying it by default understates revenue on every profitable contract.

D measures progress against the wrong denominator: `3,000,000 ÷ 10,000,000 = 30 %`, then applies that to the estimated total cost: `0.30 × 8,000,000 = 2,400,000`. Two errors compounding — comparing cost to price, then applying a progress percentage to a cost base rather than to the price.

Item 3 `[D3 · KA 3.2 · Analysis]`

A sponsor asks for a cost estimate to ±10 per cent accuracy for a scheme whose scope is defined by a concept layout and a site plan, and nothing else. The most defensible professional response is to:

- **A.** Apply a 10 per cent contingency to a bottom-up estimate to produce the range.
- **B.** Produce the ±10 per cent figure and record the assumptions it rests on.
- **C.** Build a parametric model from the concept layout and present its output at ±10 per cent.
- **D.** State that the scope definition available supports only an order-of-magnitude range.

Key: D.

Estimate accuracy is a property of the maturity of the scope definition and the estimating method used, not a service level that can be requested. The estimate classification systems in common use — those published by AACE International among them — tie accuracy ranges to the level of project definition, and a concept layout sits at the least mature end of that scale. The professional answer names what the basis will support and sets out what scope definition an estimate of the requested accuracy would require, which turns a refusal into a plan.

A confuses contingency with accuracy. Contingency is a provision for identified risk within the estimate; accuracy is a statement about the estimate's own uncertainty. Adding a percentage to the number does not narrow the range around it, and presenting it as though it did is the mechanism by which under-defined estimates become approved budgets.

B is the option most people choose, and it fails for a specific reason: recording assumptions does not repair an accuracy claim the basis cannot support. The sponsor takes the ±10 per cent into the funding paper; the assumptions stay in the appendix.

C substitutes a method for a basis. A parametric model built on concept-level inputs inherits the uncertainty of those inputs. The technique is sound and the accuracy claim is still unsupported.

Item 4 `[D4 · KA 4.2 · Application]`

Illustrative figures.

A work package budgeted 2,000 tonnes of reinforcement at USD 900 per tonne. Actual consumption was 2,100 tonnes at USD 880 per tonne. What are the price and quantity variances?

- **A.** Price USD 42,000 adverse; quantity USD 90,000 favourable.
- **B.** Price USD 90,000 adverse; quantity USD 42,000 favourable.
- **C.** Price USD 42,000 favourable; quantity USD 90,000 adverse.
- **D.** A single adverse variance of USD 48,000; the split cannot be derived from the data given.

Key: C.

```
Budget = 2,000 × 900 = 1,800,000 Actual = 2,100 × 880 = 1,848,000 Total variance = 1,848,000 − 1,800,000 = 48,000 adverse

Price variance = (budget rate − actual rate) × actual quantity = (900 − 880) × 2,100 = 42,000 favourable Quantity variance = (budget quantity − actual quantity) × budget rate = (2,000 − 2,100) × 900 = 90,000 adverse

Reconciliation: 90,000 adverse − 42,000 favourable = 48,000 adverse
```

The commercial story is the point of the decomposition: buying was good and consumption was poor, and the headline 48,000 conceals both. Reporting only the net figure loses the two facts a manager can act on.

A and **B** invert the signs or swap the labels. Both are arithmetically derived from the same two magnitudes, which is deliberate — the item tests the decomposition, not the multiplication.

D claims the split is underivable. It is derivable: the stem supplies rate and quantity for both budget and actual, which is exactly the data the decomposition needs. Candidates who select D usually know the method and have not recognised that they already have everything required.

5. Group 2 — Project management principles (Domains 5–12)

Item 5 `[D5 · KA 5.2 · Analysis]`

Illustrative figures.

The month-end cost report shows actual cost of USD 8,600,000 against a plan of USD 9,000,000, and the project manager is reporting an under-run. The cost engineer establishes that USD 1,100,000 of goods were received during the month and have not been invoiced, and that a subcontract variation instructed during the month has not yet been valued, with a first estimate of USD 400,000. The most accurate statement of the position is:

- **A.** Cost incurred is USD 9,700,000 — USD 700,000 over plan — with a further USD 400,000 trended for the variation.
- **B.** Cost is USD 8,600,000 and the project is USD 400,000 under plan; uninvoiced receipts are not yet cost.
- **C.** Cost is USD 8,600,000 and will be restated when the invoices arrive, so no action is needed now.
- **D.** Cost is USD 10,100,000, because an instructed variation is incurred as soon as it is instructed.

Key: A.

`Cost incurred = 8,600,000 + 1,100,000 = 9,700,000 Variance to plan = 9,700,000 − 9,000,000 = 700,000 over Forecast exposure including the trend = 9,700,000 + 400,000 = 10,100,000`

The item turns on the commitment–accrual–actual cycle. Goods received are cost incurred; the invoice is a document, not an economic event. An instructed but unvalued variation is a **trend** — a known exposure carried in the forecast — and not yet a measured incurred cost.

B is the position the ledger reports if nobody looks, and it is wrong by USD 1,100,000. The report measures invoices and is being read as though it measured cost.

C is **B** with a delay attached, and it is worse, because it acknowledges the problem and defers it. The whole purpose of a cut-off is that a period's cost is stated in that period.

D over-corrects. Carrying an unvalued instruction as incurred cost overstates the ledger position and breaks the reconciliation to the accounts, which is the one thing a cost report must never do. A trend belongs in the forecast, visibly, not in the actuals.

Item 6 `[D6 · KA 6.3 · Analysis]`

Illustrative figures.

A project reports budget at completion (BAC) USD 12,000,000, earned value (EV) USD 4,500,000, actual cost (AC) USD 5,000,000 and planned value (PV) USD 5,400,000. Investigation establishes that the schedule slippage was caused by a permit delay that has now been resolved, while the cost inefficiency is a systemic labour productivity shortfall expected to continue. Which estimate at completion (EAC) is the most defensible forecast?

- **A.** USD 12,000,000
- **B.** USD 12,500,000
- **C.** USD 13,333,333

- **D.** USD 15,000,000

Key: C.

`CPI = EV ÷ AC = 4,500,000 ÷ 5,000,000 = 0.90` `SPI = EV ÷ PV = 4,500,000 ÷ 5,400,000 = 0.8333`

The cost inefficiency is systemic and is expected to continue, so past cost performance should be carried forward:

`EAC = BAC ÷ CPI = 12,000,000 ÷ 0.90 = 13,333,333` (to the nearest dollar)

A is the budget at completion. It is what was agreed, not what will happen. Reporting BAC as the forecast because the baseline has not been changed is one of the most common ways a project arrives at its overrun without warning anyone.

B is `EAC = AC + (BAC − EV) = 5,000,000 + 7,500,000 = 12,500,000` — the atypical-variance form, which assumes remaining work will be performed at budgeted efficiency. It is the right method when the overrun has an identified, closed cause. Here the cause is systemic and continuing, so it understates.

D is `EAC = AC + (BAC − EV) ÷ (CPI × SPI) = 5,000,000 + 7,500,000 ÷ 0.75 = 5,000,000 + 10,000,000 = 15,000,000`, carrying both cost and schedule inefficiency forward (`0.90 × 0.8333 = 0.75`). That form is appropriate when schedule pressure is itself driving cost — overtime, acceleration, resequencing. Here the schedule driver has been removed, so applying it forecasts a penalty the project will not incur.

The assessed skill is not the arithmetic. All four figures are computable in under a minute. It is reading the situation to decide **which** inefficiency continues, and that is why the formula policy in `EXB-01 — Examination blueprint — PCL-AI` §7.2 provides no formula sheet: selecting the method is the assessment.

Item 7 `[D7 · KA 7.4 · Application]`

Illustrative figures.

A contract has a contract sum of USD 20,000,000 and provides for retention of 5 per cent, capped at 3 per cent of the contract sum. Work executed to date is valued at USD 3,400,000 gross. Previous certificates total USD 2,500,000 gross. What amount is due on this application, before tax?

- **A.** USD 730,000
- **B.** USD 855,000
- **C.** USD 900,000
- **D.** USD 3,230,000

Key: B.

First test the cap, because a candidate who does not is guessing that it does not bite:

`Retention cap = 3 % × 20,000,000 = 600,000` `Cumulative retention = 5 % × 3,400,000 = 170,000` — below the cap, so 5 per cent applies in full.

`Cumulative certified net = 3,400,000 − 170,000 = 3,230,000` `Previously certified net = 2,500,000 − (5 % × 2,500,000) = 2,500,000 − 125,000 = 2,375,000` `Due this application = 3,230,000 − 2,375,000 = 855,000`

Cross-check on the increment: `gross increment = 3,400,000 − 2,500,000 = 900,000`; `retention on the increment = 5 % × 900,000 = 45,000`; `900,000 − 45,000 = 855,000`

A deducts the cumulative retention of 170,000 from the incremental gross valuation of 900,000. Mixing a cumulative figure with an incremental one is the defining error of interim valuation, and it produces an under-certification that the contractor will find and dispute.

C ignores retention entirely and certifies the gross increment.

D is the cumulative net position — correct as a cumulative figure and wrong as an amount due, because it takes no account of what has already been certified and paid.

Item 8 `[D8 · KA 8.4 · Recall]`

The performance measurement baseline is revised:

- **A.** Whenever actual cost differs materially from planned cost.
- **B.** At each monthly reporting cycle, to reflect the latest forecast.
- **C.** When the forecast at completion exceeds the budget at completion.
- **D.** Only through approved change control, whatever the size of the variance.

Key: D.

A baseline is the agreed measurement reference. It changes only by an approved change, because variance is by definition the difference between performance and the baseline — if the baseline moves to meet performance, variance is structurally zero and the measurement has been destroyed.

A and **C** both re-baseline in response to a variance, which is precisely the circumstance in which the baseline must hold still. **B** describes rubber baselining, the practice of quietly re-setting the reference each month so that every report shows the project on plan. It is the most damaging thing that can be done to a controls function, and it is almost always done incrementally and with good intentions.

Approved change control revises a baseline for a scope, schedule or budget change that has been authorised. It does not revise a baseline because the project is behind.

Item 9 `[D9 · KA 9.5 · Application]`

Illustrative figures.

A product team of eight costs USD 120,000 per two-week sprint, fully loaded. The remaining backlog is estimated at 240 story points and is stable. The team's planned velocity is 30 points per sprint. Its observed velocities over the last five sprints were 34, 28, 31, 27 and 20 points; the most recent sprint fell across a public-holiday period. Sprints cannot be part-purchased. What is the most defensible forecast of remaining cost?

- **A.** USD 1,440,000
- **B.** USD 1,080,000
- **C.** USD 1,028,571
- **D.** USD 960,000

Key: B.

```
Average observed velocity = (34 + 28 + 31 + 27 + 20) ÷ 5 = 140 ÷ 5 = 28 points per sprint
Sprints remaining = 240 ÷ 28 = 8.57, rounded up to 9 whole sprints Remaining cost = 9 × 120,000
= 1,080,000
```

Two judgements sit inside that arithmetic, and both must be named alongside the answer. **Assumption one:** the observed five-sprint average is a better predictor than either the plan or the most recent sprint. **Assumption two:** the backlog is stable, as the stem states — if scope grows, the forecast moves, and in adaptive delivery scope growth is normal rather than exceptional.

A extrapolates from the most recent sprint alone: `240 ÷ 20 = 12 sprints`; `12 × 120,000 = 1,440,000`. The stem tells you that sprint was depressed by a holiday period. Forecasting from a single known-atypical observation is the schedule-forecasting equivalent of re-baselining to the worst month.

C treats sprint capacity as divisible: `8.57 × 120,000 = 1,028,571`. You cannot buy 0.57 of a sprint. The team is paid for whole sprints, and rounding down hides a period of cost.

D forecasts from the plan rather than from performance: `240 ÷ 30 = 8 sprints`; `8 × 120,000 = 960,000`. Substituting the plan for observed performance is the same error as reporting BAC as the forecast in Item 6, in a different vocabulary — and it is more tempting here, because the planned velocity is a published team commitment.

Item 10 `[D10 · KA 10.2 · Analysis]`

Illustrative figures.

A network has two paths from a common start to a common finish. Activity A (5 days) precedes both B (7 days) and C (3 days). C precedes D (6 days). B and D both precede E (4 days), which finishes the project. Activity B is delayed by 3 days. What is the effect?

- **A.** The project finishes one day late, and the critical path becomes A–B–E.
- **B.** No effect: activity B has two days of total float.
- **C.** The project finishes two days late, and the critical path is unchanged.
- **D.** The project finishes three days late, since the whole delay flows to the finish.

Key: A.

```
Path A–B–E = 5 + 7 + 4 = 16 days Path A–C–D–E = 5 + 3 + 6 + 4 = 18 days
```

The critical path is A–C–D–E at 18 days, and B carries total float of `18 − 16 = 2 days`.

After the 3-day delay:

```
Path A–B–E = 5 + 10 + 4 = 19 days Path A–C–D–E = 18 days (unchanged) Project duration = 19 days
— one day later than the 18-day baseline
```

Float absorbs the first two days. The third day pushes the path past 18, the project finishes one day late, and **the critical path moves to A–B–E.**

B is right up to a point and stops too early — it applies the float and does not check whether the delay exceeds it. This is the most instructive wrong answer in the set: float is a quantity that gets consumed, not a permission.

C subtracts wrongly and misses the path switch. **D** ignores float altogether and passes the whole delay to the finish date, which is what a spreadsheet does when nobody has modelled the network.

The arithmetic here is trivial by design. The item is about whether you reason over a network, and about remembering that a critical path is a property of the current schedule, not a label attached to activities at the start of the job.

Item 11 `[D11 · KA 11.3 · Analysis]`

In a four-person project office, the same cost engineer raises purchase requisitions, records goods receipts in the enterprise resource planning system, and approves supplier invoices for payment up to a threshold. The control weakness this creates is best described as:

- **A.** A data-quality risk, since one person's coding conventions are never independently checked.
- **B.** A capacity risk, since the cost engineer becomes a single point of failure during absence.
- **C.** One person controls initiation, receipt and approval, so a fictitious purchase could be paid unseen.
- **D.** An acceptable efficiency in a small team, provided the approval threshold is set low.

Key: C.

Segregation of duties exists so that no single person can both initiate and complete a transaction that moves money. Here the same individual controls all three stages of the procure-to-pay cycle, so a fictitious or inflated purchase could be requisitioned, receipted and approved without any second party seeing it. That is the control weakness; everything else is a consequence or an unrelated risk.

A and **B** describe genuine risks that this arrangement also creates. Neither is the control weakness, and a candidate who selects either has identified a real problem and answered a different question.

D is the option that most deserves examining, because it is half right in a way that matters in practice. Perfect segregation is genuinely impossible in a four-person office, and pretending otherwise produces a control framework nobody follows. But a low threshold is a **compensating control**, not the absence of a weakness — and a compensating control only counts when it is documented, operated and monitored. The professional response to a small team is to name the weakness, then put in place compensating controls that can be evidenced: independent review of a sample of transactions, and exception reporting that lists every case where requisition, receipt and approval share a user identity. Accepting the arrangement without either is the answer D actually describes.

Item 12 `[D12 · KA 12.2 · Analysis]`

Illustrative figures.

A single risk, if it occurs, would delay handover by four months and trigger liquidated damages of USD 4,000,000. The probability is assessed at 15 per cent. The finance director proposes carrying USD 600,000 of contingency for it, being `0.15 × 4,000,000`. The most defensible controls response is to:

- **A.** Carry USD 600,000 as contingency, since expected monetary value is the standard basis for quantifying risk.
- **B.** Carry USD 4,000,000 as contingency, since the outcome is binary and the project must absorb it.
- **C.** Carry nothing, since liquidated damages are a contractual remedy and not a project cost risk.
- **D.** Report the exposure as 0 or USD 4,000,000 at 15 per cent likelihood, and set the reserve by recorded management decision.

Key: D.

`Expected monetary value = 0.15 × 4,000,000 = 600,000` — and the project will never experience USD 600,000 of this risk. It will experience nothing or USD 4,000,000.

Expected monetary value is a long-run average. It is a sound basis for aggregating a portfolio of many independent risks, where the averaging has something to average over. Applied to a single dominant binary risk it produces a number that is arithmetically correct and describes no possible outcome, and it conceals the decision that actually has to be taken: how much of a 15 per cent chance of a USD 4,000,000 hit the organisation is willing to carry unfunded. That is a risk-appetite decision for management, and the controls professional's job is to present the exposure so that it can be taken knowingly and recorded.

A is not an arithmetic error; it is a modelling error, and it is extremely common because the calculation is easy and the number looks responsible in a report.

B is not wrong in principle — fully funding a severe binary exposure is a legitimate choice. It is not the most defensible response, because it is a decision the controls professional does not have the standing to take unilaterally, and because reserving the full amount by default freezes capital that the organisation may rationally prefer to put at risk. This is a single-best-answer item in the strict sense: **B** is permissible, **D** is what a competent professional does first.

C excludes a quantifiable cost consequence on the grounds that its origin is contractual. Liquidated damages hit the project's cost outturn like any other cost.

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6. Group 3 — AI knowledge & practical approach (Domain 13)

Item 13 `[D13 · KA 13.1 · Recall]`

Which statement about the output of a large language model (LLM) is correct?

- **A.** It is retrieved from a verified source, so it can be cited directly.
- **B.** It is generated as a likely continuation, so it can be fluent and wrong.
- **C.** It is deterministic: an identical prompt always returns identical text.
- **D.** It carries a reliability score indicating whether the content is correct.

Key: B.

A large language model produces text by generating statistically likely continuations, not by retrieving checked statements from a store. Fluency and correctness are therefore **independent properties of the output** — which is the single most consequential fact for any professional who relies on one, because human readers use fluency as a proxy for competence and the proxy does not hold here.

A describes retrieval. A generative model can produce a citation-shaped string that refers to nothing at all, and it will produce it with exactly the same confidence as a real one.

C is false in general. Most deployments sample from the distribution, so repeated identical prompts can return different text — a genuine problem for auditability, and one reason a deterministic rule is preferred wherever the logic is known (see Item 15). Even where sampling is disabled, reproducibility is not correctness.

D conflates two different quantities. Some interfaces expose token-level probabilities, but the probability that a token is a likely continuation is not the probability that a statement is true. A confident model and a correct model are not the same thing.

Item 14 `[D13 · KA 13.2 · Analysis]`

A controls team wants to train a model to predict cost overrun at control-account level using five years of historical project data. The cost ledger has been recoded twice in that period, and neither earlier coding structure has been mapped to the current one. The most important consequence is:

- **A.** Training will take longer, because the model must reconcile three coding structures.
- **B.** The model will favour recent projects, which is corrected by adding more historical data.
- **C.** Features and labels mean different things across the period, so the model learns from incompatible definitions.
- **D.** The model cannot be built until the whole ledger history has been recoded to the current structure.

Key: C.

The same field name refers to different things in different years. A model trained across that history learns from a mixture of incompatible definitions, and — this is the part that makes it dangerous rather than merely disappointing — **a held-out test set drawn from the same inconsistent history will report respectable accuracy.** The failure is silent. It surfaces only when the model is used on live data and produces confident predictions that nobody can explain.

A treats a validity problem as a performance problem. Training time is irrelevant.

B misdiagnoses the mechanism and then prescribes more of the poison: adding further data from the same inconsistently coded history does not repair a definition problem, it enlarges it.

D overstates. Recoding the entire history is one remedy and rarely the practical one. Restricting training to the consistently coded period, or building an explicit mapping between structures and validating it on a sample, are both legitimate. The professional response is to establish what the data means before modelling it, and to document that establishment — which is the whole of knowledge area 13.2 in one sentence.

Item 15 `[D13 · KA 13.4 · Analysis]`

Each week, a controls team wants to flag every control account whose cost performance index has fallen by more than a published threshold since the previous period, and to issue a short written explanation of each flag. The best-governed division of labour is:

- **A.** A deterministic rule detects the deterioration; a generative model drafts the explanation; the engineer verifies both.
- **B.** A generative model both detects the deterioration against the threshold and drafts the explanation.
- **C.** A generative model detects the deterioration and the cost engineer writes each explanation.
- **D.** A model is trained on which accounts cost engineers have flagged before, and flags accordingly.

Key: A.

Match the technique to the task. Detection here is a **defined, auditable rule**: compute an index, compare the change against a published threshold, flag. That is deterministic logic, it is reproducible, and it can be reconciled — run it twice and it gives the same answer, and anyone can check why a particular account was flagged. Explanation is a **language task**, where a generative model genuinely helps, provided a competent human owns the output before it is issued. The verification step is not a formality; it is the step that makes the whole arrangement defensible.

B hands a deterministic test to a probabilistic system. It buys opacity and non-reproducibility for nothing: two runs may disagree, and the flag list cannot be reconciled to the underlying figures. Whenever a rule will do, a rule is better — cheaper, faster, auditable and explainable.

C allocates both tasks to the participant worst suited to each: the model to the reproducible test, the human to the repetitive drafting. It is the mirror image of the right answer.

D solves a problem nobody has. Training a model to reproduce what cost engineers have flagged before encodes past human behaviour, including its inconsistencies and its blind spots, in place of a published threshold that already exists and that anyone can read. It also makes the flag list impossible to explain to a sponsor, since the answer to "why was this account flagged?" becomes "because the model learned that we usually flag accounts like this".

Item 16 `[D13 · KA 13.6 · Analysis]`

A generative-AI assistant drafts the cost narrative for the monthly report. The cost manager reviews it, corrects two figures, and issues the report. For the audit trail, the minimum that must be recorded is:

- **A.** A statement in the report that AI tools were used in its preparation.
- **B.** Nothing: the cost manager issued the report and is accountable for it.
- **C.** A per-paragraph confidence score reported by the model alongside the draft.
- **D.** The prompt, the model, the output as generated, the corrections and the approver.

Key: D.

Auditability means being able to reconstruct, afterwards, what the tool produced, what a human changed, and who owned the result. All four elements are needed: without the output as generated you cannot see what was corrected, and without the approver you cannot see who took responsibility.

A is a disclaimer, not a trail. It tells a reader that a tool was involved and nothing about what was checked, which is precisely the information a reviewer needs.

B is the option that deserves the most attention, because its premise is correct and its conclusion is wrong. Accountability does sit with the cost manager — "AI proposes; the professional disposes" is not negotiable. But accountability that cannot be evidenced cannot be demonstrated when it is questioned six months later, and demonstrating it is the entire purpose of an audit trail. A professional who is accountable and has no record is exposed, not protected.

C substitutes a machine's self-report for human review. A high confidence score attached to a fluent paragraph is still an unverified paragraph, and confidence is not correctness (see Item 13).

7. Answer key

Item	Key	Item	Key	Item	Key	Item	Key
1	A	5	A	9	B	13	B
2	B	6	C	10	A	14	C
3	D	7	B	11	C	15	A
4	C	8	D	12	D	16	D

8. Reading the items as an item writer

Publishing the rationale is useful. Publishing what the item is doing is more useful, and almost nobody does it. Four observations, keyed to the items above.

Every distractor is an arrival point, not a decoy. In Item 9, each of the three wrong answers is the figure you actually get by making one specific decision: extrapolating from a single atypical sprint, treating sprint capacity as divisible, or forecasting from the plan instead of from performance. A candidate who lands on one of them has not been tricked — they have completed a calculation, and the option they selected identifies exactly which judgement they made. That is what makes the pattern of a candidate's wrong answers diagnostic, and it is the standard `EXB-04 — How items are written, reviewed and retired` §4 holds authors to.

The arithmetic is deliberately easy where the judgement is hard. In Item 6, all four estimates at completion are computable in a minute; in Item 10, the path sums are trivial. The difficulty is entirely in deciding which inefficiency continues, and in remembering that float is consumed. An item that hides its assessment behind long multiplication is measuring stamina.

The key must be the best answer, not the only permissible one. Item 12 is the clearest case: fully reserving a severe binary exposure is a legitimate choice, and the item says so in the rationale rather than pretending option B is foolish. Candidates who argue with such an item instead of answering it lose the mark; candidates who ask "which of these would I do first" find it.

Option length is a cue, and judgement items fight it. The defensible answer in an Analysis item usually needs more words than a wrong one, because it carries a qualification the wrong answers do not. Item 12's key is the longest option in that set, which is a weakness the reviewers accepted rather than concealed — the alternative was to pad the distractors, which would have made three options less plausible to buy symmetry. Where you can see the tension in these items, you are seeing an honest trade-off rather than an oversight, and a candidate who has learned to notice option length has learned something that will help them on a badly written examination somewhere else.

9. How candidates get these wrong

Reading the options before establishing the position. The single most expensive habit. Items 5, 6 and 10 all become straightforward once you have written down what the data actually says, and all three become a guess between plausible alternatives if you have not.

Doing the arithmetic and stopping. Item 6 punishes this precisely: computing all four estimates at completion correctly and then choosing without deciding which inefficiency continues.

Answering the question you know instead of the one asked. Item 11 offers two real risks that are not the control weakness. Item 7 offers a figure that is correct as a cumulative position and wrong as an amount due.

Applying a method outside its assumptions. Item 12 is expected monetary value used where averaging has nothing to average over. Item 3 is an accuracy claim the basis cannot support.

Treating "the professional answer" as the cautious answer. It is not always. In Item 12 the cautious option — reserve the full amount — is not the key, because the decision is not the controls professional's to take alone.

Accepting a machine output because it reads well. Items 13, 14 and 16 all turn on the same point in different vocabulary. Fluency, confidence and accuracy are three different things, and only the third matters.

10. Checklist for working practice items

- [ ] I answered before reading the rationale, and wrote down my intermediate figure or position.
- [ ] For every item I got wrong, I identified which distractor I chose and which named error it represents.
- [ ] For every item I got right, I checked my reasoning against the rationale rather than my letter against the key.
- [ ] I worked at least one set under time pressure rather than at leisure.
- [ ] I noted which domains my errors cluster in, and adjusted study time using the weighting arithmetic in `EXB-02` §3.
- [ ] I have not tried to memorise these items, which are study material and will not appear on any form.

Publishing sixteen items with every distractor explained gives away nothing an examination needs to keep, and it settles a question a candidate would otherwise have to take on trust: whether "scenario-based" and "assesses judgement" describe the items or the marketing. Work them, and the claim in `EXB-01 — Examination blueprint — PCL-AI` becomes something you can check rather than something you are asked to believe.

Related

- `EXB-01 — Examination blueprint — PCL-AI` — the sampling frame these items are written against, and the formula policy Item 6 depends on
- `EXB-03 — Cognitive levels and item types` — what Recall, Application and Analysis mean, and the limits of the format
- `EXB-07 — Examination security and integrity` — why published items and live items are permanently separate
- `CER-01 — Certification handbook — master` — eligibility, booking, conduct and results

Sources and standards

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